Zhuoran Wang

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DOB: 18 August 1995
Nationality: Chinese

Summary

Interested in and experienced in microwave photonics and integrated photonics

Education

Bachelor's degree: 09.2013 - 07.2017
Harbin Engineering university, optoelectronic information science and engineering

Master's degree: 09.2017 - 08.2020
Huazhong University of Science and Technology, optical engineering

Doctoral degree: 09.2021 - 12.2022 (Dropped)
University of Ottawa, electronic and computer science

Doctoral degree: 09.2023 – 06.2026
McGill University, electronic and computer science

Work Experience

Accelink Technologies (01.2023 - 07.2023): Optical Engineer
- Designed integrated optical gyroscopes.
- Troubleshot and resolved technical issues in fiber-based optical gyroscope products.

Huazhong University of Science and Technology (08.2020 - 04.2021): Research Assistant
- Designed the finite element simulation system for simulating bi-anisotropic waveguide

Research Publications

Accepted Journal Articles

1. **Zhuoran Wang**, Santiago Bernal, David V. Plant, Lawrence R. Chen, "Breaking the Phase Noise–Mode Purity Trade-Off in Long-Loop OEOs via Cascaded Zero-Dispersion Recirculating Loops," *Light: Science and Applications*, **Accepted** (2026).

Published Journal Articles

1. **Zhuoran Wang**, Santiago Bernal, David V. Plant, Lawrence R. Chen, " Ultra-High-Order Harmonic Mode-Locked Optoelectronic Oscillator," *Laser and Photonics Reviews*, in print (2026). https://doi.org/10.1002/lpor.202502273
2. **Zhuoran Wang**, Santiago Bernal, David V. Plant, Lawrence R. Chen, " Dispersion-tuned Mode-locked Optoelectronic Oscillator," *Optics Letters*, 51(9), 2416-2419 (2026).
3. **Zhuoran Wang**, Yuxuan Xie, Zixian Wei, Jinsong Zhang, David V. Plant and Lawrence R. Chen, "Highly Reconfigurable Narrowband Microwave Photonic Filter Based on Chirp-Like Sliced ASE Source," *IEEE Transactions on Microwave Theory and Techniques*, 73(10), 7973-7980 (2025).
4. **Zhuoran Wang**, *et al.*, "Versatile Microwave Photonic Signal Processor Based on a Quantum-Dash Mode-Locked Laser," *Journal of Lightwave Technology*, 43(21), 9842-9849, (2025).
5. **Zhuoran Wang**, Santiago Bernal, David V. Plant, Lawrence R. Chen, “Independently Tunable Dual-Frequency Optoelectronic Oscillator with Single-Mode Operation Via Cascaded Zero-Dispersion Recirculating Loops”, *Journal of Lightwave Technology*, 44(2), 567-574 (2025).
6. **Zhuoran Wang**, Yuxuan Xie, Zixian Wei, David V. Plant and Lawrence R. Chen, "Versatile Optoelectronic Oscillator for Single-Frequency, Dual-Frequency, and Mode-Locking Oscillation," *Journal of Lightwave Technology*, 43(14), 6554-6562 (2025).
7. Yixuan Liu, Yangxue Ma, **Zhuoran Wang**, Zhen Zeng, Ziwei Xu, Lingjie Zhang, Yaowen Zhang, Zhiyao Zhang, Heping Li, and Yong Liu, "Microwave pulse generation with independently tunable pulse width and repetition rate in an actively mode-locked optoelectronic oscillator driven by an amplitude-modulated signal," *Optics Express*, 33(18), 37381-37393 (2025).
8. Zheng Dai, **Zhuoran Wang**, Jianping Yao, “Dual-loop parity-time symmetric system with a rational loop length ratio”, *Optics Letters*, 48(1), 143-146 (2022).
9. Xiong Zhongfei, Chen Weijin, **Zhuoran Wang**, Xu Jin and Yuntian Chen, “Finite element modeling of electromagnetic properties in photonic bianisotropic structures,” *Frontiers of Optoelectronics*, 14(2), 148-153 (2021).
10. Zhongfei Xiong, Qingdong Yang, Weijin Chen, **Zhuoran Wang**, Jing Xu, Wei Liu, and Yuntian Chen, "On the constraints of electromagnetic multipoles for symmetric scatterers: eigenmode analysis," *Optics Express*, 28(3), 3073-3085 (2020).
11. Bei Wu, **Zhuoran Wang(co-author)**, Weijin Chen, Zhongfei Xiong, Jing Xu, and Yuntian Chen, "S-parameters, non-Hermitian ports and the finite-element implementation in photonic devices with $\mathcal{PT}$-symmetry," *Optics Express*, 27(13), 17648-17657 (2019).

Conference Proceedings

1. **Zhuoran Wang**, Santiago Bernal, David V. Plant, Lawrence R. Chen, “ASE-Driven OEO with Cascaded Zero-Dispersion Recirculating Loops Unlocks 458th-order Harmonic Mode-Locking”, in *2025 International Topical Meeting on Microwave Photonics (MWP)*, Quebec, Canada, 2025 **(Post-deadline Paper).**
2. **Zhuoran Wang**, Santiago Bernal, David V. Plant, Lawrence R. Chen, “Dispersion-tuned Mode-locked Optoelectronic Oscillator”, in *2025 International Topical Meeting on Microwave Photonics (MWP)*, Quebec, Canada, 2025 **(Nominated Best Student Paper).**
3. **Zhuoran Wang**, Santiago Bernal, David V. Plant, Lawrence R. Chen, “Single-mode Optoelectronic Oscillator Based on ASE Source and Cascaded Zero-dispersion Recirculating Loops”, in *2025 International Topical Meeting on Microwave Photonics (MWP)*, Quebec, Canada, 2025.

4. **Zhuoran Wang**, Santiago Bernal, David V. Plant, Lawrence R. Chen, "Improving ASE-based Optoelectronic Oscillator Performance Using a Chirp-like Slicing Approach", in *2025 International Topical Meeting on Microwave Photonics (MWP)*, Quebec, Canada, 2025.
5. **Zhuoran Wang**, Yuxuan Xie, Zixian Wei, David V. Plant, Lawrence R. Chen, "Tunable Single and Dual-Frequency Optoelectronic Oscillator Incorporating a Single Cavity", in *2024 International Topical Meeting on Microwave Photonics (MWP)*, Pisa, Italy, 2024, pp. 1-4
6. **Zhuoran Wang**, Lawrence R. Chen, "Compact TE-pass Polarizer Covering Full Communication Band with Ultra-high Extinction Ratio in O-band and C+ L band", in *Advanced Photonics Congress 2024*, Technical Digest Series (Optica Publishing Group, 2024), paper ITu2B.4.

Patents

1. Yuntian Chen(supervisor), **Zhuoran Wang**. A simulation method and system of Bi anisotropic waveguide, China, CN112231947B [P]. 2020-09-16.

Industry Collaboration Projects

07.2024 - 03.2026 **Quasi-coherent Optical Communication using Quantum-dash Mode-locked Laser** (McGill University in collaboration with Fonex Data Systems/Bilfrost Communications/National Research Council Canada)

- Collaborate with FONEX engineers to test and evaluate quasi-coherent optical communication link
- Designed quasi-coherent optical communication +QD-MLL link architecture, built the set-up, measured and analyzed experimental results.

04.2025 - 08.2025 **On-chip MMI design** (Internship at Fonex Data Systems)

- Designed 3dB MMI, arbitrary splitting ratio MMI, anti-reflection MMI, and 90° hybrid MMI on SOI
- Characterized the fabricated MMIs using a PIC testing platform

SKILLS

Languages: 7 in IELTS test

Simulation software: Lumerical, Comsol, Opti-system, VPI Photonics, HFSS

Coding: Matlab, Python, Mathematica.

Layout software: Klayout, L-edit, Blender

Awards & Scholarships

1、 Douglas Van Graduate Impact Award from McGill University (2026); C$2,000.

2、 Intern Scholarship, Mitacs (2025) — Developing high performance on-chip couplers at FONEX; C$ 15,000.

3、 Shining Star Travel Grant, IEEE Canada Students & Young Professionals Congress (2025) — Travel support for a high-quality poster presentation; C$455 total (C$255 registration + C$200 travel).

4、 1st Prize, COPL Annual Conference — Photonic Systems & Microwave Track (2025) — Voted first prize (~1/10); C$300.

5、 1st Prize, STARaCom Student Poster Competition (2025) — First place (1/30); C$1,000.

6、 Merit Scholarship from The University of Ottawa (2022); C$4,000.

7、 China Scholarship Council (CSC) Doctoral Scholarship (2021–2022) — Awarded for PhD studies at the University of Ottawa; total value C$105,600 for 4 years. However, I quit the PhD at the end of year 2022.

8、 Zhixing Award, Huazhong University of Science and Technology (2019) — Recognized for outstanding academic and practical performance; C$300.